

Student's last name and name:

De Simone Antonio

Two exercises on the linear theory

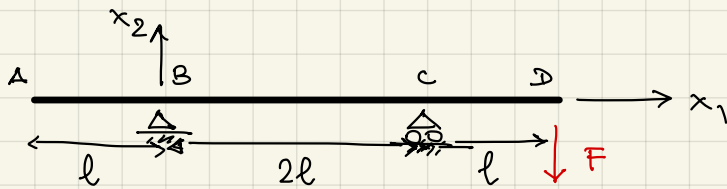
Using linear theory, compute and graph the internal forces, and compute reactions and elastic displacements in the cases sketched below, write answers on this sheet

Exercise 1

(total 150 points)

Data: $\underline{F} = -F \underline{e}_2$ applied in D

Constraints: $\underline{x}(B) = \underline{0}$
 $\underline{x}(C) \cdot \underline{e}_2 = 0$



$$R_{B1} = 0$$

$$R_{B2} = -\frac{1}{2} F$$

$$R_{C2} = \frac{3}{2} F$$

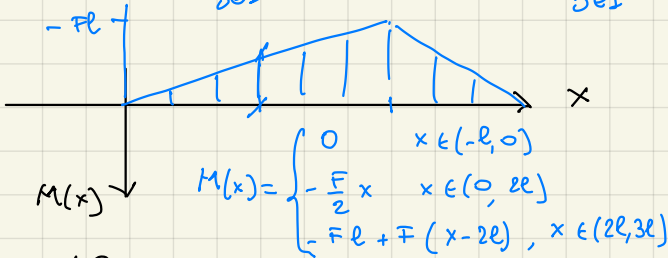
(10 points each)

$$\varphi_A = w'(-l) = \frac{Fl^2}{3EI}$$

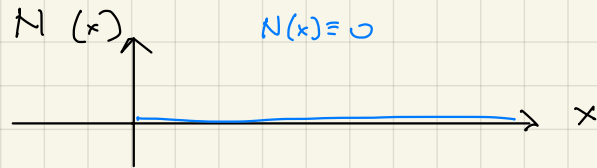
$$\varphi_B = w'(0) = \frac{Fl^2}{3EI}$$

$$\varphi_C = w'(2l) = -\frac{2Fl^2}{3EI}$$

(20 points each)



Diagrams M, N, T
(20 points each)

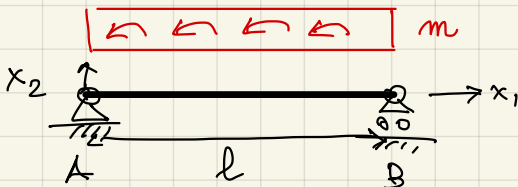


Exercise 2

(total 130 points)

Data: m constant distributed couple

Constraints: $\underline{x}(A) = \underline{0}$
 $\underline{x}(C) \cdot \underline{e}_2 = 0$



$$R_{A1} = 0$$

$$R_{A2} = m$$

$$R_{B2} = -m$$

(10 points each)

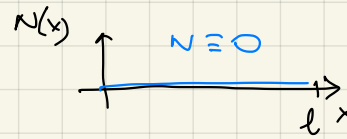
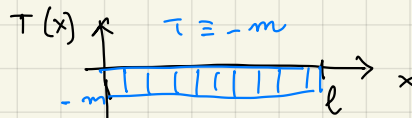
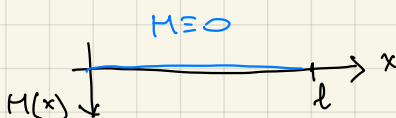
$$\varphi_A = w'(0) = 0$$

$$\varphi_B = w'(l) = 0$$

$$w\left(\frac{l}{2}\right) = 0$$

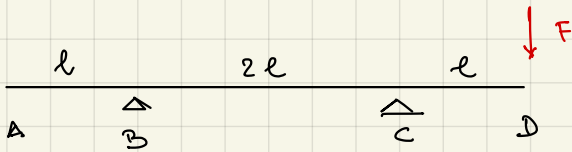
$$u(l) = 0$$

(10 points each)



Diagrams M, N, T
(20 points each)

Ex. 1



ISO

FBD



$$R_{B1} = 0$$

$$R_{B2} + R_{C2} - F = 0$$

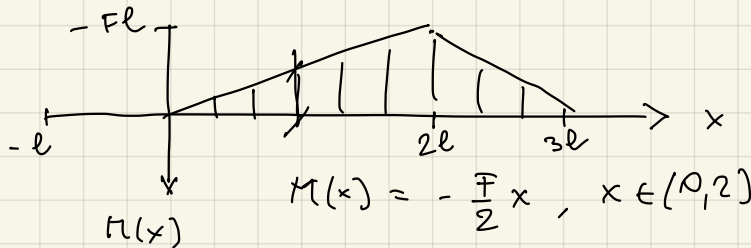
$$\Rightarrow R_{B2} = F - \frac{3}{2}F = -\frac{F}{2}$$

$$-F \cdot 3l + R_{C2} \cdot 2l = 0 \Rightarrow R_{C2} = \frac{3}{2}F$$

$$N(x) \equiv 0$$

$$T(x) = \begin{cases} 0 & x \in (-l, 0) \\ F/2 & x \in (0, 2l) \\ -F & x \in (2l, 3l) \end{cases}$$

$$M'' = f_2 = 0 \Rightarrow M(x) \text{ piecewise affine}$$



$$w'' = \frac{M}{EI}$$

$$x \in (0, 2l) \quad w''(x) = -\frac{F}{2EI} x$$

$$w'(x) - w'(0) = -\frac{F}{4EI} x^2 \Rightarrow w'(x) = w'(0) - \frac{F}{4EI} x^2$$

$$w(x) = \cancel{w(0)} + \cancel{w'(0)}x - \frac{F}{12EI} x^3$$

$$w(2l) = w'(0) \cdot 2l - \frac{F \cdot 8l^3}{12EI} = 0 \Rightarrow w'(0) = \frac{Fl^2}{3EI} = \varphi_B$$

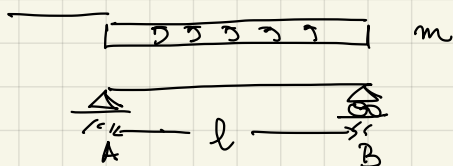
$$\Rightarrow w(x) = \frac{Fl^2}{3EI} x - \frac{F}{12EI} x^3$$

$$[\text{Check: } w(2l) = \frac{2Fl^3}{3EI} - \frac{F \cdot 8l^3}{12EI} = 0 \checkmark]$$

$$\varphi_B = w'(0) = \frac{Fl^2}{3EI}, \quad \varphi_C = w'(2l) = \frac{Fl^2}{3EI} - \frac{F \cdot 4l^2}{4EI} = -\frac{2}{3} \frac{Fl^2}{EI}$$

$$\varphi_A = \varphi_B = \frac{Fl^2}{3EI} \text{ because } w''(x) = \frac{M}{EI} = 0 \text{ for } x \in (-l, 0), \text{ so } w' = \text{const in } (-l, 0).$$

Ex. 2



ISO



$$R_{A1} = 0$$

$$R_{A2} + R_{B2} = 0$$

$$\Rightarrow R_{A2} = m$$

$$R_{B2}l + ml = 0 \Rightarrow R_{B2} = -m$$

$$T' + f_2 = 0 \Rightarrow T(x) \equiv -m = \text{const.}$$

$$M' + T + \underbrace{m}_{-m} = M' = 0 \Rightarrow M(x) = \text{const} = M(0) = 0$$

$$v'' = \frac{M}{EI} \equiv 0 \Rightarrow w(x) \text{ affine, and } \equiv 0 \text{ because } w(0) = w(l) = 0$$

$$\varphi(x) = w'(x) \equiv 0$$

$$N(x) \equiv 0 \Rightarrow u'(x) = \frac{N}{EA} = 0 \Rightarrow u(x) \equiv u(0) = 0$$

Histogram of Given Data (Bin width = 50)

